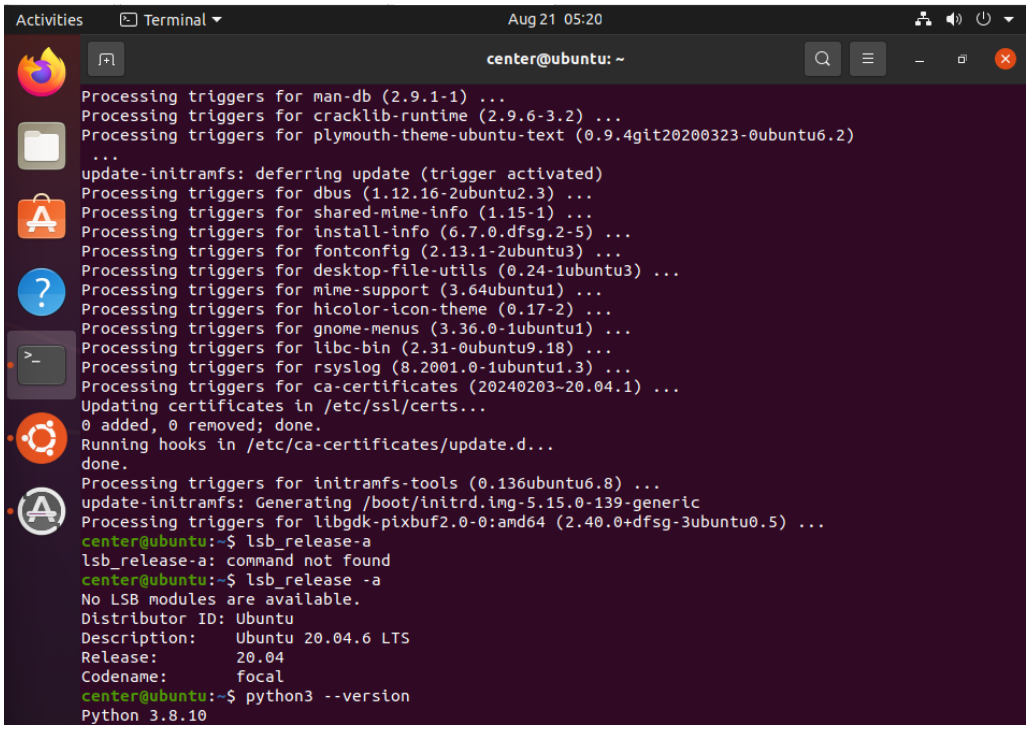
977-337 Network Programming and Simulation

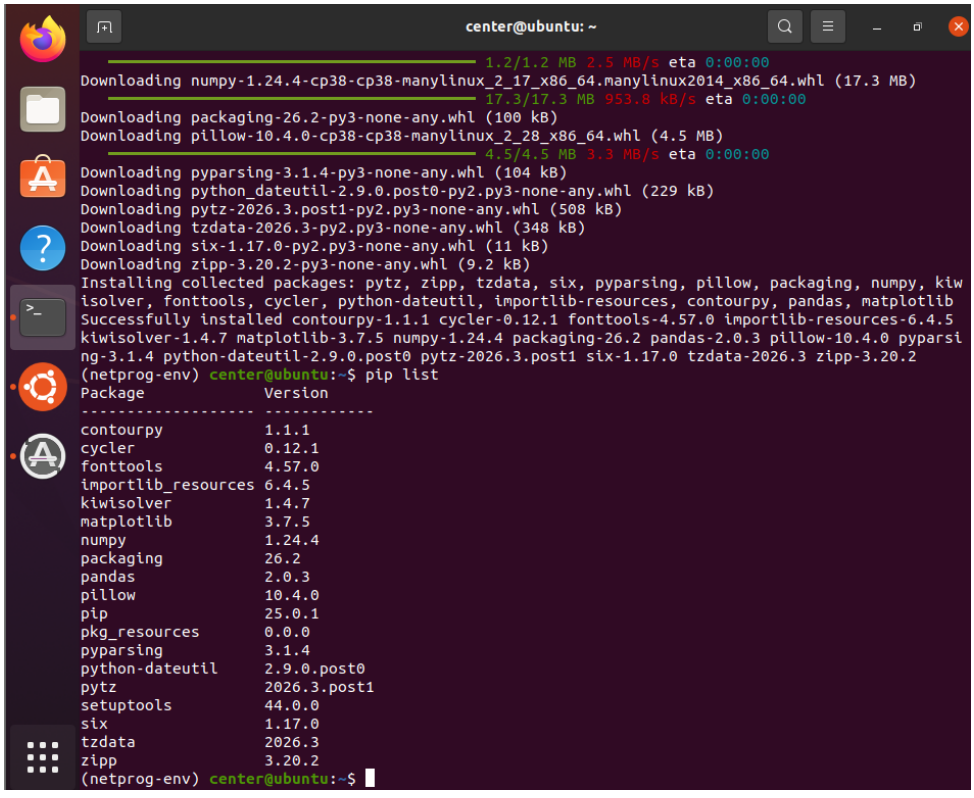
Lab 1

Environment Setup: Linux, Python, Git, Mininet & Mininet-WiFi

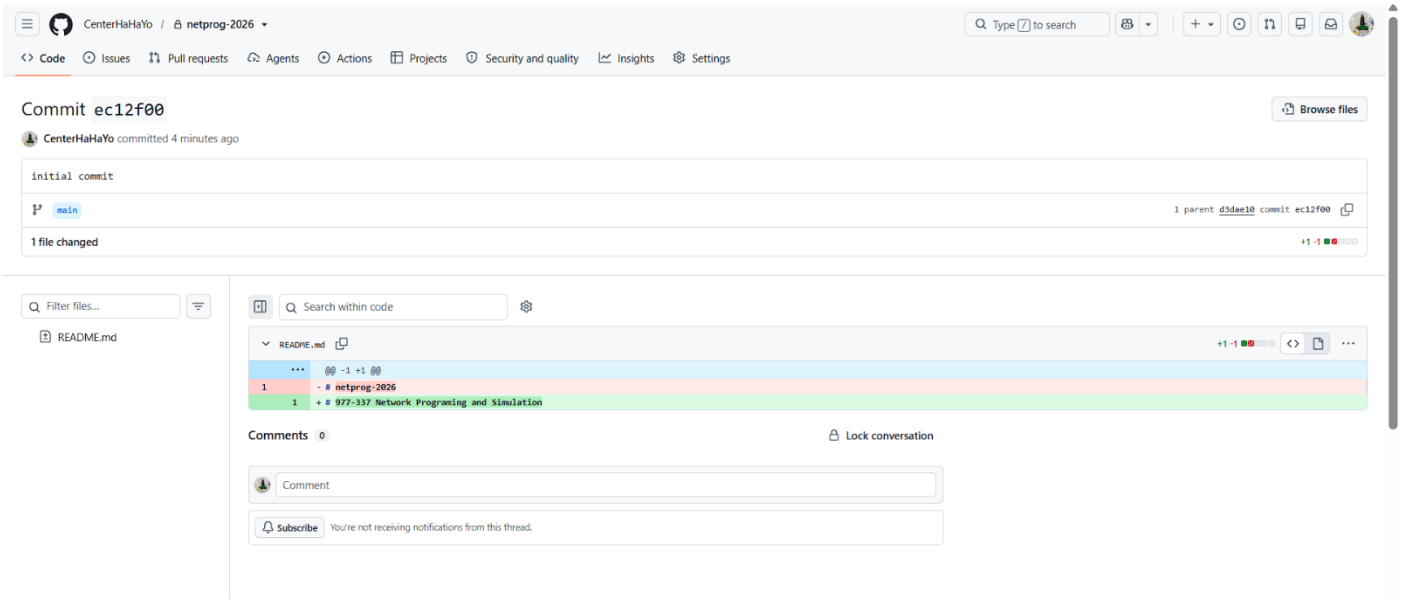
Task 1: Provide a screenshot of lsb_release -aand python3 --version confirming your OS and Python versions



Task 2: Provide a screenshot of your active netprog-env virtual environment prompt, and the output of pip list showing matplotlib and pandas installed



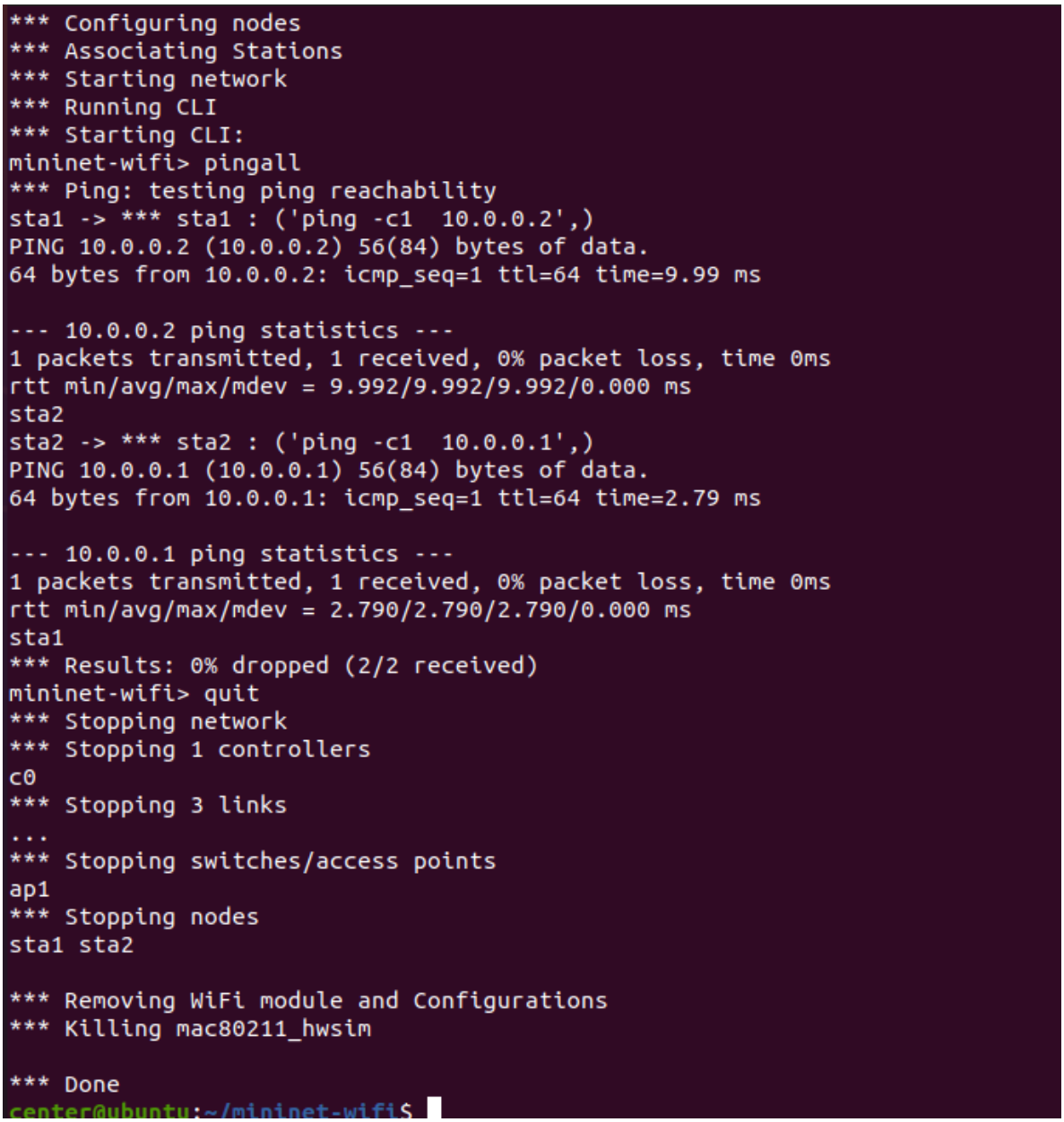
Task 3: Provide the URL of your GitHub repository (netprog-2026) and a screenshot of its initial commit history.



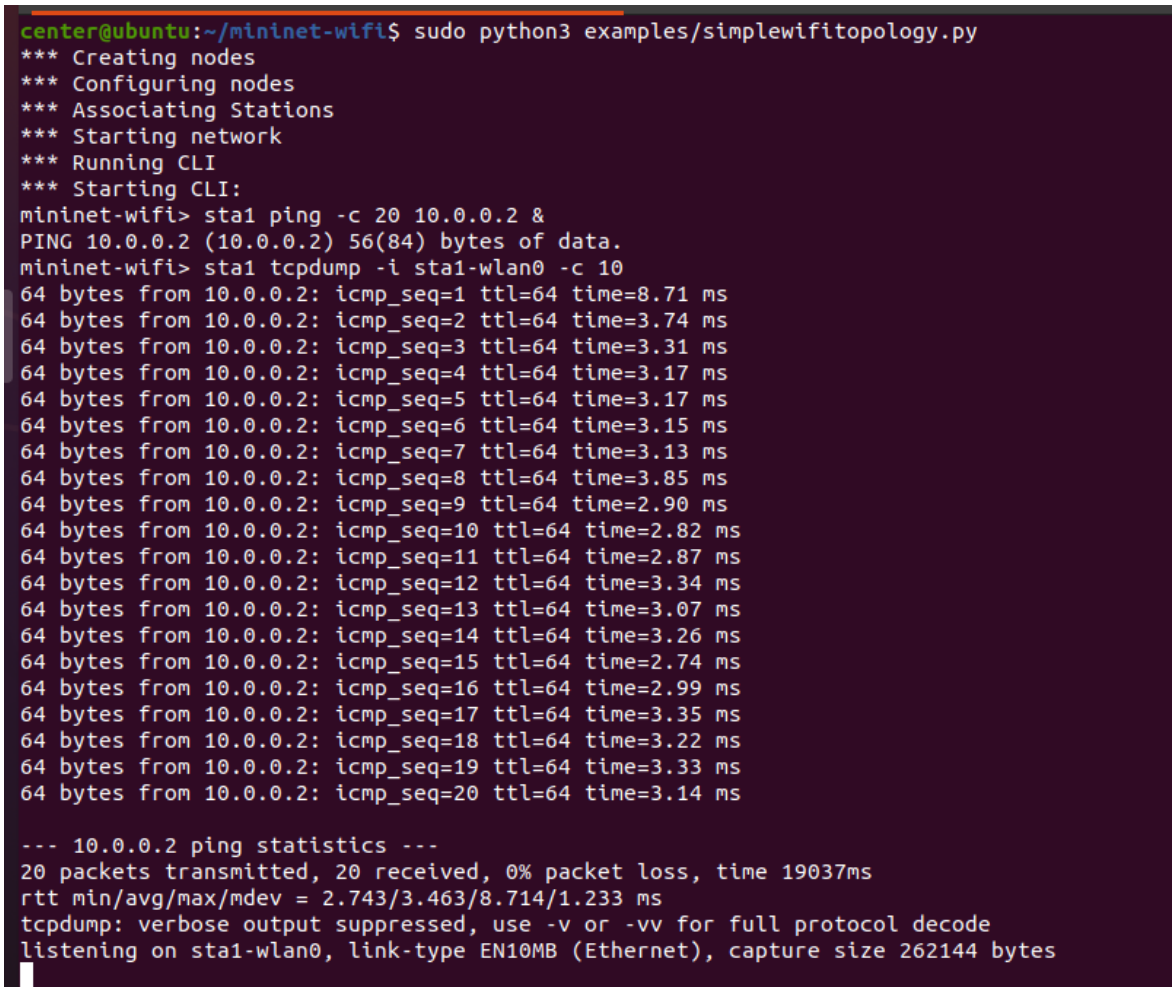
Task 4: Provide a screenshot of sudo mn --test pingall showing 0% dropped packets.

```
center@ubuntu:~/mininet$ sudo mn --test pingall
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
completed in 5.384 seconds
center@ubuntu:~/mininet$
```

Task 5: Provide a screenshot of the Mininet-WiFi sample topology running, including the pingall result.



Task 6: Provide a screenshot of the tcpdump capture from Part F, and in 2-3 sentences, explain in your own words why this confirms Mininet-WiFi is emulating (not purely simulating) the wireless link.



Reflection Questions

1.) In your own words, what is the difference between network simulation and network emulation? Which one is Mininet-WiFi, and how did Part F demonstrate this?

Network simulation uses network behavior mathematical techniques or algorithms without running actual operating system-level networking code, but network emulation uses real networking components and protocols in a virtual environment. Mininet-WiFi is a network emulator. Part F's tcpdump capture showed actual packets being generated and captured on the station interface, demonstrating that real network traffic was emulated, not purely simulated.

2.) Why does the course ask you to use a Python virtual environment (netprog-env) instead of installing packages directly with pip at the system level?

Because netprog-env is a Python virtual environment so that the packages are kept separate from the system Python and other projects. This helps prevent package conflicts and makes it easier to manage the packages.

3.) What was the most difficult part of the installation process for you, and how did you resolve it?

The most difficult part for me had many parts for example, in part E I use sudo util/install.sh -Wlnfv. It was filed because I may forget to use pip install --upgrade pip in netprog-env lead to run sudo python3 mininet-wifi/examples/simplewifi.py command was filed. So, I deleted it using sudo, then clone mininet-wifi again. And when I ran sudo python3 mininet-wifi/examples/simplewifi.py, the result was can not open file and no such file or directory. So, I ran sudo python3 examples/simplewifitopology.py instead.